

Parameters (millions)		d_model	ffw_size	kv_size	n_heads	n_layers
This work	Chinchilla					
7	-	128	512	32	4	3
14	-	224	896	32	7	4
20	-	288	1152	32	7	5
38	-	448	1792	32	7	6
52	44	512	2048	64	8	8
66	57	576	2304	64	9	9
83	74	640	2560	64	10	10
97	90	640	2560	64	10	13
112	106	640	2560	64	10	16
125	117	768	3072	64	12	12
146	140	768	3072	64	12	15
168	163	768	3072	64	12	18
182	175	896	3584	64	14	14
201	196	896	3584	64	14	16
220	217	896	3584	64	14	18
255	251	1024	4096	64	16	16
280	278	1024	4096	64	16	18
305	306	1024	4096	64	16	20
421	425	1280	5120	128	10	18
480	489	1280	5120	128	10	21
502	509	1408	5632	128	11	18
539	552	1280	5120	128	10	24
574	587	1408	5632	128	11	21
619	632	1536	6144	128	12	19
645	664	1408	5632	128	11	24
704	724	1536	6144	128	12	22
789	816	1536	6144	128	12	25
865	893	1792	7168	128	14	20
981	1018	1792	7168	128	14	23
1096	1143	1792	7168	128	14	26
1215	1266	2048	8192	128	16	22
1364	1424	2176	8704	128	17	22
1366	1429	2048	8192	128	16	25
1517	1593	2048	8192	128	16	28
1535	1609	2176	8704	128	17	25
1650	1731	2304	9216	128	18	24
1706	1794	2176	8704	128	17	28
1905	2007	2304	9216	128	18	28
2160	2283	2304	9216	128	18	32
2179	2298	2560	10240	128	20	26
2494	2639	2560	10240	128	20	30
2809	2980	2560	10240	128	20	34
3090	-	2688	10752	128	22	34
3263	3530	2688	10752	128	21	36
3574	3802	2816	11264	128	22	36
3900	4084	2944	11776	128	23	36
4239	4516	3072	12288	128	24	36
6355	6796	3584	14336	128	28	40
8672	9293	4096	16384	128	32	42
10912	11452	4352	17408	128	32	47
11455	12295	4608	18432	128	36	44
12220	12569	4608	18432	128	32	47
13601	13735	4864	19456	128	32	47
14917	14940	4992	19968	128	32	49
15056	16183	5120	20480	128	40	47

Table 15: **Model architectures.** We list the architectures of all models trained as part of this work. Many shown models have been trained multiple times on different amounts of unique data and for varying epochs.

T Prompts and Samples

The following figures illustrate the prompts with samples from each evaluation dataset. Prompts stem from PromptSource [5] or GPT-3 [15]. All data comes from the ground truth datasets in this section, and no generations are shown here.

Context →	Edmond (or Edmund) Halley, FRS (pronounced ; 8 November [O.S. 29 October] 1656 - 25 January 1742 [O.S. 14 January 1741]) was an English astronomer, geophysicist, mathematician, meteorologist, and physicist who is best known for computing the orbit of Halley's Comet. He was the second Astronomer Royal in Britain, succeeding John Flamsteed. Question: Edmond Halley was born outside of the United Kingdom. True, False, or Neither? Answer:
Correct Answer →	Neither
Incorrect Answer →	True
Incorrect Answer →	False

Figure 21: Formatted dataset example from ANLI R1 evaluated using accuracy as described in [Appendix K](#).

Context →	The 1970 Swedish Open was a combined men's and women's tennis tournament played on outdoor clay courts held in Båstad, Sweden and was part of the Grand Prix circuit of the 1970 Tour. It was the 23rd edition of the tournament and was held from 2 July through 12 July 1970. Dick Crealy and Peaches Bartkowicz won the singles titles. Question: Dick Crealy and Peaches Bartkowicz beat eachother in the 1970 Swedish Open. True, False, or Neither? Answer:
Correct Answer →	False
Incorrect Answer →	True
Incorrect Answer →	Neither

Figure 22: Formatted dataset example from ANLI R2 evaluated using accuracy as described in [Appendix K](#).

Context →	Tokyo - Food group Nestle is seeking to lure Japanese holiday shoppers with a taste for fine snacking with a gold-wrapped Kit Kat chocolate bar. The single finger Kit Kat is wrapped in a thin layer of gold leaf. Only 500 of the bars go on sale from Dec. 29 with a price tag of around 2,016 yen (\$16). The Kit Kat chocolate bar made its debut in Japan in 1973 and since then a variety of flavors - from green tea to wasabi - have been produced. Question: Japanese like kit kat. True, False, or Neither? Answer:
Correct Answer →	True
Incorrect Answer →	False
Incorrect Answer →	Neither

Figure 23: Formatted dataset example from ANLI R3 evaluated using accuracy as described in [Appendix K](#).

Context →	An astronomer observes that a planet rotates faster after a meteorite impact. Which is the most likely effect of this increase in rotation?
Correct Answer →	Planetary days will become shorter.
Incorrect Answer →	Planetary years will become longer.
Incorrect Answer →	Planetary gravity will become stronger.

Figure 24: Formatted dataset example from ARC-Challenge evaluated using accuracy as described in [Appendix K](#).

Context →	To express the distance between the Milky Way galaxy and other galaxies, the most appropriate unit of measurement is the
Correct Answer →	light-year.
Incorrect Answer →	meter.
Incorrect Answer →	kilometer.
Incorrect Answer →	astronomical unit.

Figure 25: Formatted dataset example from ARC-Easy evaluated using accuracy as described in [Appendix K](#).

Context →	Radio wave - Radio waves are a type of electromagnetic radiation with wavelengths in the electromagnetic spectrum longer than infrared light. Radio waves have frequencies as high as 300 gigahertz (GHz) to as low as 30 hertz (Hz). At 300 GHz, the corresponding wavelength is 1 mm, and at 30 Hz is 10,000 km. Like all other electromagnetic waves, radio waves travel at the speed of light. They are generated by electric charges undergoing acceleration, such as time varying electric currents. Naturally occurring radio waves are emitted by lightning and astronomical objects. Question: do radio waves travel at the speed of light? Answer:
Correct Answer →	yes
Incorrect Answer →	no

Figure 26: Formatted dataset example from BoolQ evaluated using accuracy as described in [Appendix K](#).

Context →	A: Okay. So Frank, what, uh, type of, uh, budget do you or your family have? B: Well, uh I don't know that we really have a budget. Question: he and his family really have a budget. True, False or Neither? Answer:
Correct Answer →	False
Incorrect Answer →	True
Incorrect Answer →	Neither

Figure 27: Formatted dataset example from CB evaluated using accuracy as described in [Appendix K](#).

Context →	The computer was expensive to fix therefore
Correct Answer →	I bought a new one.
Incorrect Answer →	I got it repaired.

Figure 28: Formatted dataset example from COPA evaluated using accuracy as described in [Appendix K](#).